

Executive Summary

This report tests a trading strategy called pairs trading on two chip companies, NVIDIA (NVDA) and Broadcom (AVGO). It uses five years of daily share prices. The first three years were used to check whether the two share prices tend to move together. The last two years (roughly October 2024 to September 2026) were used to see how the strategy would have performed on data it had not seen before.

The strategy did not work. Over the two-year test, it lost 26.55%. Whereas simply buying and holding both shares in equal amounts gained 98.51%. Only one of its seven trades made money, and three trades were closed early by a safety limit (a stop-loss). The evidence that the two shares move together was weak to begin with. Moreover, the strategy's way of balancing the two sides of each trade became unreliable, which left some trades with very little protection. We see that a trading signal is not enough on its own. The two shares need a stable, lasting relationship for this kind of strategy to work.

Pairs Trading Methodology

If two shares usually move together, then when they drift apart, one is probably “too expensive” compared with the other, and the gap should close again. The strategy bets on the gap closing by buying the cheaper share and short selling the more expensive one. If the gap closes, the strategy makes money whichever way the overall market moves.

Three steps turn this idea into rules:

- **Hedge ratio (β):** how much AVGO to trade against each NVDA share. It is re-estimated every day using the previous 60 trading days of prices (about three months), so it can adapt as conditions change.
- **Spread:** the gap between the two prices after adjusting for the hedge ratio. It is calculated as the NVDA price minus ($\beta \times$ the AVGO price).
- **Z-score:** a measure of how unusual today's spread is compared with its own last 60 days. A Z-score of 0 is normal, +2 is unusually high, and -2 is unusually low.

Each day's signal only uses information available at that day's closing price, and trades happen at that close. This means the strategy never uses information from the future.

Strategy Description and Strategic Rationale

Pairs Strategy – NVDA and AVGO

Both companies are large chip makers that depend on the same boom in artificial intelligence (AI) spending, so it is reasonable to expect their share prices to move together. The trading rules are:

- **Entry (spread unusually low):** if the Z-score falls below -2, buy NVDA and short AVGO.
- **Entry (spread unusually high):** if the Z-score rises above +2, short NVDA and buy AVGO.
- **Take profit:** close the trade when the Z-score returns to 0, meaning the gap has closed.
- **Stop-loss:** close the trade immediately and accept the loss if the Z-score reaches

+3.5 or - 3.5, meaning the gap is getting wider instead of closing.

After a stop-loss, the strategy waits for the Z-score to return to a normal range before trading again. This avoids repeatedly re-entering a losing situation.

Benchmark – 50/50 Buy and Hold

To judge the results, the strategy is compared with a simple “buy and hold” portfolio, with half the money in each share, bought at the start and never touched. This is a tough comparison because it fully benefits from the strong rise in chip shares, whereas pairs trading aims to profit from the gap between two shares, not from the market rising. A fairer minimum standard for the pairs strategy is to beat a return of zero (keeping the money in cash).

Key Assumptions and Parameter Selection

The backtest (a test of the strategy on past data) uses the following settings, which were chosen in advance and not adjusted to improve the results:

- **Data split:** the first three years were used to test whether the shares move together, and the last two years were used for trading.
- **Window:** 60 trading days for both the hedge ratio and the Z-score.
- **Trading rules:** enter at a Z-score of ± 2 , take profit at 0, and stop-loss at ± 3.5 .
- **Costs:** 0.05% (5 basis points) of the trade value each time a trade is opened or closed. Cash earns nothing.

Choosing the settings in advance avoids “overfitting”, which means tweaking a strategy until it looks good on past data. However, it also means the results show only one version of the strategy. While in a trade, the strategy invests all of its money across the two shares according to the hedge ratio. When it is not in a trade, it holds cash. The analysis ignores some real-world costs, such as fees for borrowing shares to short, taxes and the effect of large trades on prices, and it uses nominal returns (not adjusted for inflation).

Key Terms

- **Cointegration:** when two prices can wander but tend to stay in a stable relationship over time, like a person walking a dog on a lead.
 - **ADF test and p-value:** a statistical test used to check whether the gap between two prices keeps returning to normal. A p-value below 0.05 is the usual standard for being confident that it does.
 - **Sharpe ratio:** the return earned for each unit of risk taken. Above 1 is generally considered good, and below 0 means the strategy lost money.
 - **Maximum drawdown:** the largest fall from a previous peak to a later low point.
 - **Win rate:** the share of trades that made money.
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Backtest Results and Analysis

Do the Two Shares Move Together? (In-Sample Test)

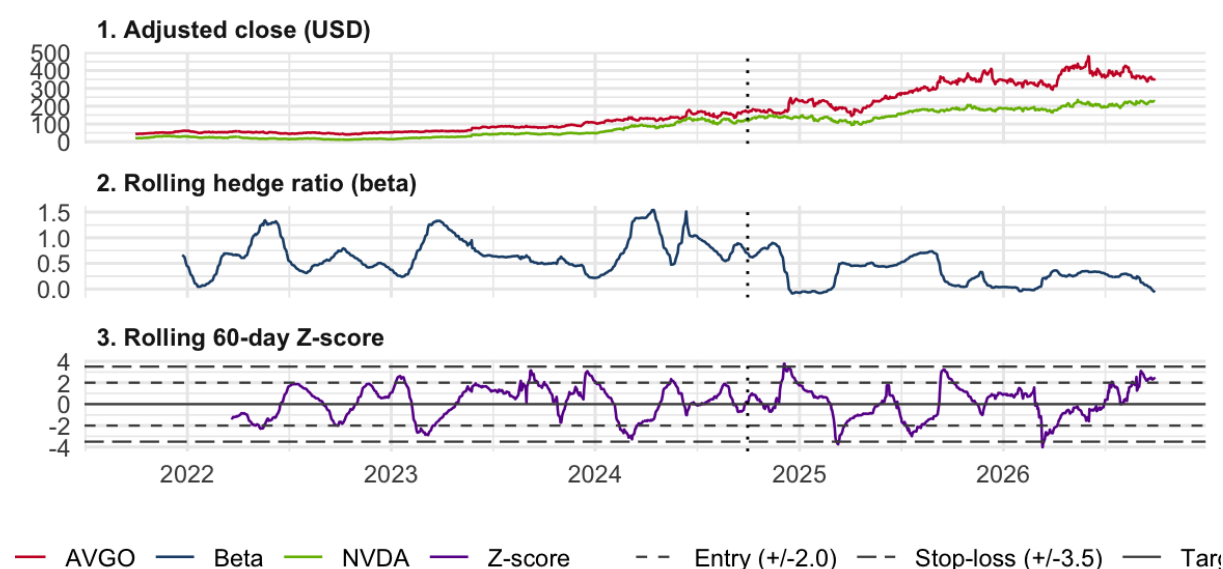
Before trading, the strategy needed evidence that NVDA and AVGO really do move together. Using the first three years of data, the test looked at the gap between NVDA's price and the price predicted from AVGO (using a fixed hedge ratio of 0.89). If the two shares are linked, this gap should keep returning to normal. The result was an ADF statistic of -3.347 and a p-value of 0.0625. This is borderline. The p-value is above the usual 0.05 cut-off, and the statistic sits almost exactly on the threshold (-3.34) used for this type of test. In plain terms, there was some sign of a link, but not enough to be confident about it.

Dynamic Spread Diagnostics

The chart below shows the share prices (top), the hedge ratio (middle) and the Z-score (bottom). The dashed lines on the bottom panel mark the entry (± 2), take-profit (0) and stop-loss (± 3.5) levels. The dotted vertical line marks the start of the test period.

NVDA / AVGO: Dynamic Spread Diagnostics

Dotted vertical line = start of out-of-sample period



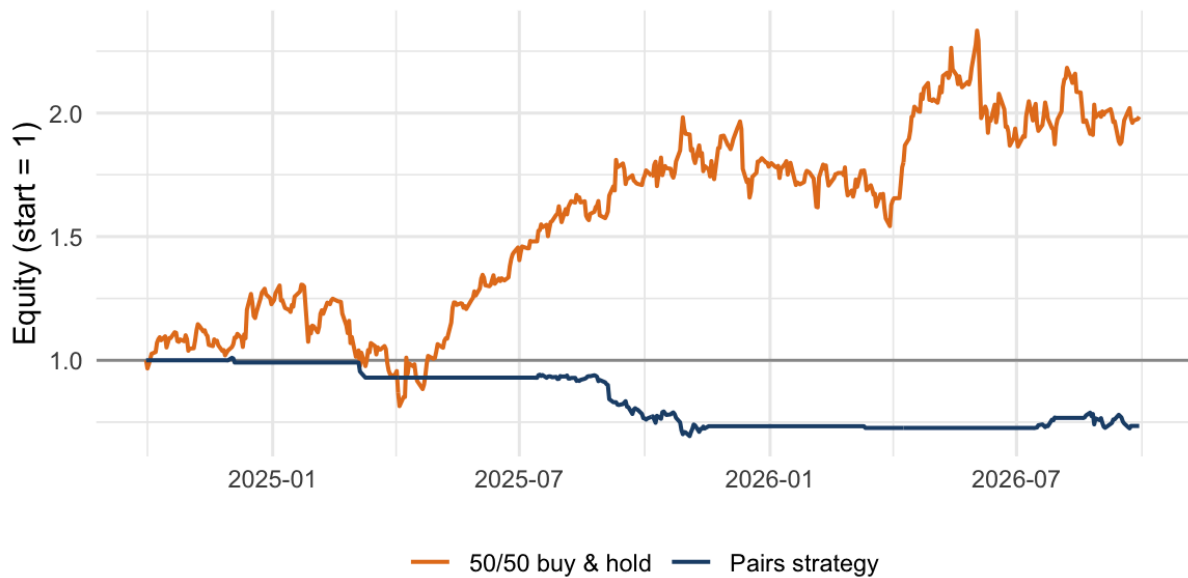
The key problem is the hedge ratio, which should stay fairly steady if the two shares have a lasting relationship. In the first three years, it swung between about 0.05 and 1.5. In the test period, it fell to roughly zero in early 2025, recovered to about 0.7 in late 2025, and fell to near zero again in 2026 (slightly below zero at the very end). When the hedge ratio is near zero, the strategy is no longer really trading one share against the other. It puts almost all of its money in NVDA, so the trade becomes a bet on NVDA alone, with little protection against NVDA's strong upward trend. The Z-score also reached the ± 3.5 stop-loss level three times in the test period, at around December 2024, March 2025 and March 2026.

Out-of-Sample Performance Attribution

The graph below shows how one unit of money (for example, \$1) would have grown under the pairs strategy and under buy and hold, after transaction costs.

Out-of-Sample Performance Attribution

Growth of 1 unit of capital (net of transaction costs)



The strategy's line is flat for long stretches because it was not in a trade. It also shows two big falls: about 6 percentage points in early 2025, and a slow slide of about 20 points from mid-2025 to late 2025. A small recovery in mid-2026 left it at about 0.73 of its starting value. Buy and hold ended near 2.0, roughly doubling the original money.

Metric	Strategy	50/50 Buy & Hold
Total Return	-26.55%	98.51%
Annualised Sharpe Ratio	-1.11	0.95
Maximum Drawdown	-31.34%	-37.60%
Win Rate (%)	14.3	n/a
Total Trades	7	n/a
Stop-Loss Triggers (+/-3.5)	3	n/a
Long-spread stops ($z \leq -3.5$)	2	n/a
Short-spread stops ($z \geq +3.5$)	1	n/a
Profit-Target Exits ($z \rightarrow 0$)	3	n/a
Closed at End of Sample	0	n/a
Forced Exits (beta ≤ 0 / invalid data)	1	n/a

Performance Analysis

- Return and Sharpe ratio:** The strategy lost 26.55%, while buy and hold gained 98.51%. The Sharpe ratio was -1.11 for the strategy, meaning it lost money for the risk taken, compared with a solid 0.95 for buy and hold.

- **Drawdown:** The strategy's largest fall from a peak was 31.34%, compared with 37.60% for buy and hold. This looks better, but the strategy's fall came on top of losses, while the benchmark's fall was a temporary dip within a large overall gain.
- **Trade results:** Only 1 of 7 trades (14.3%) made money. Three trades were closed by the stop-loss (two when the spread was unusually low, one when it was unusually high). Three reached the take-profit level, and one was closed because the hedge ratio dropped to zero or below, which makes no sense for a pairs trade.

Since only one trade made money, at least two of the three "take-profit" trades still ended in a loss. A likely explanation is that the Z-score compares the spread with its recent 60-day average. If the spread settles at a new level, the average catches up and the Z-score returns to zero even though the gap never actually closed. The strategy then exits while the trade is still losing.

Risk Assessment

The stop-loss limited the damage from each trade, but it could not fix a weak strategy. It was triggered in three of the seven trades, much more often than expected if the gap between the shares reliably returned to normal. Suggesting the relationship between the shares changed during the test period.

There are other risks. The hedge ratio was unstable, so the amount of protection kept changing. The strategy also put all of its money to work each time it traded, and did not trade smaller when markets were more volatile. With only seven trades, one or two events can swing the result, so no firm conclusions can be drawn about how the strategy would perform in the future. Real-world costs, such as borrowing fees for the short side, would make the results worse.

Conclusion and Investment Implications

This backtest does not support trading NVDA against AVGO using these rules. Evidence that the shares move together was borderline, the hedge ratio was unstable, and the strategy lost 26.55% while the shares roughly doubled. This applies to this specific version of the strategy, which uses raw prices and a 60-day window, and not to pairs trading in general.

Possible improvements include measuring the relationship using percentage returns instead of raw prices, using a longer window or a smoother method (such as a Kalman filter) so the hedge ratio swings less, only trading when the relationship passes a statistical test, adding a time limit for trades that are not working, trading smaller when markets are volatile, and testing other pairs of shares. More broadly, two companies in the same industry are not automatically a good pair. Statistical evidence of a stable relationship is necessary.

Sources:

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